

GENE 8160: ADVANCED EVOLUTIONARY GENETICS

Spring 2010, 3 credits

Class meetings: Tuesday and Thursday 9:30-10:15 am, Life Sciences Room B116

Prerequisites: GENE 8150 or GENE 6000

Instructors: Kelly Dyer and John Wares

Contact Information and Office Hours

KD: Email: kdyer@uga.edu, Phone: 706-542-3154, Office: Life Sciences C226

Office hours: Tuesday and Thursday 10:15 – 11 am and by appointment (please email)

JW: Email jpwares@uga.edu, Phone: 706-542-7720, Office: Life Sciences C326

Office hours: By appointment (please email)

Course description and goals

This course explores the process of evolution at the molecular level. We will use basic evolutionary principles to explore how DNA changes through time; to infer population and species history from DNA sequences; to determine what forces have shaped the evolution of genes and genomes; and to understand the relationship between molecular evolution and phenotypic evolution. This field is large and diverse, and combines population genetics, classical genetics, phylogenetics, molecular biology, and genomics. The goals of this class are to introduce you to some of the main questions, give you a feel for the kinds of data used, and expose you to the tools that are used to answer questions.

Texts and assigned readings

We will use the following book as a basic reference:

Principles of Population Genetics, 4th edition (2008) by D. L. Hartl and A. G. Clark.

Each topic will be supplemented with articles and reviews from the primary literature. Articles will be available as PDF files on eLearning Commons.

Other useful resources (most are at the library)

Population Genetics, A concise guide, 2nd ed. J. Gillespie. Johns Hopkins Press

Fundamentals of Molecular Evolution, 2nd ed. By D. Grauer and W.-H. Li. Sinauer Press.

Computational Molecular Evolution, Z. Yang. Oxford University Press.

Class format, Participation, and Attendance

This class will combine lectures, discussions, and computer activities. We expect that you come to class prepared and ready to engage in whatever activities we will be doing that day.

Participation is essential for lively discussion, and your contribution to this is reflected in the participation portion of your grade. Regular attendance in lecture is expected and necessary for learning the material, which is cumulative in nature. After two unexcused absences, each additional absence will result in the loss of 1% from your final average, except in the case of a documented medical or extreme personal emergency.

Grading

Grades are A-F and will be based on three exams (20% each), assignments and presentations (30%), and participation (10%).

Lecture Handouts

A pdf copy of the lecture slides will be available before class on eLearning Commons.

Exams

Three non-cumulative exams will be given during lecture, on 2/9/10, 3/23/10, and 4/27/10. Makeup exams will only be given in the case of a documented medical or extreme personal emergency. All make-up exams will be oral examinations.

Short assignments

Short assignments may come in many flavors, depending on the nature of the material. In addition to traditional problem sets, throughout the semester you will write many short 'response' papers (~ 1 page typed) to journal articles or class discussions. For journal articles, these should contain (1) the main question is being addressed, (2) what the authors did, (3) what the authors found, (4) the significance of the findings, and (5) a question you had about the paper (e.g. What left you baffled? What techniques were new? Were the conclusions inappropriate? Was the design inadequate?) and (6) two to three questions or topics for discussion.

Response papers on journal articles are due by 5 pm the day before class.

Discussions

Frequently we will spend part of a class period discussing a paper from the primary literature. When you present a paper, you will introduce the paper in a 10-15 minute presentation and then lead the discussion of the paper. You will likely want to come prepared with some 'kick off' questions for discussion. You will be evaluated based on if you introduced the reading in a knowledgeable manner, presented data/figures/results clearly and concisely, engaged the class in insightful discussion, effectively answered questions, and in general helped to explain the science. You need to know the paper you are presenting backward and forward and inside out, and you probably want to read related papers as well.

Electronics

Please turn your phone off during class and office hours.

Religious Holidays

If you plan to miss a class because of a religious holiday, you must talk to one of the instructors at least fourteen days prior to the class to make arrangements to make-up missed work.

Influenza

The University Health Center asks that students, faculty or staff who live either on or off campus and have an influenza-like illness remain at home until at least 24 hours after they are free of fever (100° F [37.8°C]) or signs of a fever without the use of fever-reducing medications. Please do not come to class if you suspect you might have the flu. More information is available at: <http://www.uhs.uga.edu/healthtopics/influenza.html>.

Honor Code

All students will abide by the UGA Student Honor Code: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." All suspected incidences of academic dishonesty, as outlined under prohibited conduct in the UGA academic honesty policy (<http://www.uga.edu/ovpi/honesty/sect05.htm>), will be treated following UGA guidelines (http://www.uga.edu/ovpi/honesty/culture_honesty.htm).

Disclaimer

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

GENE 8160: Tentative course schedule

Class	Date		Tentative Topic
1	Th 1/7	JW	Life history I
2	Tu 1/12	JW	Life history II
3	Th 1/14	JW	Effective population size I
4	Tu 1/19	JW	Effective population size II
5	Th 1/21	JW	Molecular population genetics I
6	Tu 1/26	JW	Molecular population genetics II
7	Th 2/28	JW	Molecular population genetics III
8	Tu 2/2	JW	Phylogeography I
9	Th 2/4	JW	Phylogeography II
10	Tu 2/9	JW	EXAM #1
11	Th 2/11	KD	Population genomics
12	Tu 2/16	KD	Detecting Selection I
13	Th 2/18	KD	Detecting Selection II
14	Tu 2/23	KD	Detecting Selection III
15	Th 2/25	KD	Detecting Selection IV
16	Tu 3/2	KD	Phylogenetic methods I
17	Th 3/4	KD	Phylogenetic methods II
	Tu 3/9		Spring Break: No class
	Th 3/11		Spring Break: No class
18	Tu 3/16	KD	Phylogenetic methods III
19	Th 3/18	KD	Phylogenetic methods IV
20	Tu 3/23	KD	EXAM #2
21	Th 3/25	KD	Genome evolution I
22	Tu 3/30	KD	Genome evolution II
23	Th 4/1	KD	Gene duplication I
24	Tu 4/6	KD	Gene duplication II
25	Th 4/8	KD	Sexual Selection and sex chromosomes I
26	Tu 4/13	KD	Sexual Selection and sex chromosomes II
27	Th 4/15	KD	Organelles and endosymbiosis
28	Tu 4/20	KD	Gene regulatory evolution I
29	Th 4/22	KD	Gene regulatory evolution II
30	Tu 4/27	KD	EXAM #3